

Bilakshana Neupane

AI, ML & Full-Stack
Development



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- Kathmandu, Nepal
- 2007/03/02
- Nepali
- Female
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Education

Bachelor of Science in Computer Science

King's College/Westcliff University

2024 – present | Kathmandu, Nepal

Currently pursuing a Bachelor of Science in Computer Science (BSCS) at Kings College, focusing on programming, software development, and computer systems.

+2 in Management

Hetauda School of Management

Hetauda, Nepal

Completed +2 in Management from Hetauda School of Management, gaining basic knowledge of business, accounting, and management skills.

Skills

Python



Core syntax, OOP, data analysis, web frameworks, scripting, and API integration.

C



Core syntax, pointers, memory management, data structures, file handling, and debugging.

Data Science



Data cleaning, statistical analysis, data visualization, machine learning, and SQL.

Artificial Intelligence



Machine learning, deep learning, neural networks, natural language processing (NLP), computer vision, TensorFlow/PyTorch, and model evaluation.

C++



Proficient in C++ including OOP, STL, data structures, memory management, file I/O, templates, exception handling, multithreading basics, and modern C++ features.

Django



Web Development



Interests

Coding

Strong problem-solving skills through coding challenges

Reading

Regularly follow technology blogs and research papers

Drawing

Courses

Data Sciences and ML with Python

Broadway Infosys

Kathmandu, Nepal

Completed a Data Science with Python training course from Broadway Infosys, Nepal, covering data analysis, visualization, machine learning, and Python libraries like Pandas and NumPy. Gained practical skills in handling real-world datasets and building predictive models.

Languages

English

IELTS

Nepali

Hindi

Spanish

Projects

Breast Cancer Prediction Web App [🔗](#)

End-to-End ML Pipeline with Streamlit Deployment

Developed a full machine learning pipeline using Support Vector Machine (SVM) to predict breast cancer from cell data. Included data preprocessing, visualization, model training, and evaluation. Deployed the model using Streamlit for real-time, user-friendly predictions.

Amazon Alexa Reviews Sentiment Analysis [🔗](#)

Python, Machine Learning, Natural Language Processing, Streamlit

Built a machine learning model to classify Amazon Alexa product reviews as positive or negative using Natural Language Processing (NLP) techniques. Applied models like Random Forest and XGBoost, and deployed the project using Streamlit for interactive use.

Titanic Survival Prediction [🔗](#)

Applied Machine Learning with the Titanic Dataset

Designed a comprehensive machine learning pipeline to predict Titanic passenger survival using the Kaggle Titanic dataset. Covered the full workflow from data preprocessing and visualization to model training, evaluation, and prediction.

MindfulJournal [🔗](#)

Flask-Based Journaling App for Daily Motivation and Mindfulness

MindfulJournal is a beautifully designed Flask web application that allows users to record daily affirmations, motivations, and positive thoughts. It features a sleek user interface, real-time entry visualization, and floating animations—offering a mindful, stylish journaling experience.

Artificial Intelligence Training

Broadway Infosys

Kathmandu, Nepal

Completed an Artificial Intelligence with Python training program focused on machine learning, deep learning, and AI model development. Gained hands-on experience with Python-based AI libraries, data preprocessing, model training, and implementing intelligent solutions. Developed practical skills in building predictive models and applying AI techniques to real-world problems.

Amazon Product Data Scraper

Python Script for Automated Web Scraping and Data Collection

Developed a Python script to scrape product details from Amazon search results, including title, price, rating, review count, and availability. The extracted data is stored in a CSV file for easy analysis and reporting.

Cell Sample Classification Using SVM

SVM-Based Benign vs. Malignant Prediction with Data Visualization

Implemented a Support Vector Machine (SVM) classifier using scikit-learn to predict whether cell samples are benign or malignant from cytology data. Included data visualization with matplotlib to interpret model results.

Life Expectancy Prediction

Data Science Pipeline with EDA and Interactive Streamlit App

Executed a full data science project involving exploratory data analysis (EDA) and machine learning to predict life expectancy based on health and economic indicators. Developed an interactive Streamlit web app for real-time predictions and data visualization using Python, pandas, scikit-learn, and Streamlit.

Banking Management System (C Programming)

Console-Based Banking Application with File Handling

Developed a menu-driven banking management system in C that enables account creation, deposits, withdrawals, balance checks, and account closure. Implemented file handling for persistent data storage, demonstrating core programming concepts through practical banking operations.

Smart Waste Image Classifier

Python, Deep Learning, Computer Vision, TensorFlow, Streamlit

Built an AI-based smart waste image classification system using Convolutional Neural Networks (CNN) and transfer learning models like MobileNetV2 and ResNet50. Applied image preprocessing and augmentation techniques to classify waste into categories such as plastic, paper, metal, glass, and organic. Developed an interactive Streamlit application for real-time waste classification and prediction.

IMDb Movie Review Sentiment Analysis System

Python, Natural Language Processing (NLP), Machine Learning, TF-IDF, Logistic Regression, Streamlit

Developed a Natural Language Processing (NLP) based sentiment analysis system to classify IMDb movie reviews as positive or negative. Applied text preprocessing techniques, TF-IDF vectorization, and Logistic Regression for sentiment classification. Evaluated model performance using accuracy, F1-score, ROC-AUC, and confusion matrix visualization. Deployed the system using Streamlit with features like real-time text prediction and batch CSV analysis.

Domain-Specific RAG Chatbot

Python, Generative AI, RAG, LLaMA 3.1, LangChain, FAISS, Sentence Transformers, Streamlit

Developed a Retrieval-Augmented Generation (RAG) chatbot that provides accurate, document-based answers using Large Language Models (LLMs). Implemented document processing, text chunking, embedding generation, and FAISS-based vector search to retrieve relevant information from domain-specific PDFs and text files. Integrated LLaMA 3.1 through Groq API for generating grounded responses and deployed an interactive Streamlit interface for document-based question answering.